

i) trace Quick sort as shown in the class for the following Array.

i) 85 76 99 13 45 103 -5 26 31

ii) 350 790 400 200 100 96 900 890 760

Ans

last =		first =		pivot = 0	
i	j	a[i]	a[j]	a[pivot]	
0	8				

85	76	99	13	45	103	-5	26	31
0	1	2	3	4	5	6	7	8

pivot is first element in array 85

$$a[i] \leq a[\text{pivot}]$$

1) $85 \leq 85$ T $i++$

2) $76 \leq 85$ T $i++$

3) $99 \leq 85$ F

$$a[j] \geq a[\text{pivot}]$$

$31 \geq 85$ F

pivot = 85

swap i and j

swap 99 and 31

0 1 2 3 4 5 6 7 8 $i=2$
 $j=8$

85 76 31 13 45 103 -5 26 99

$$a[i] \leq a[\text{pivot}]$$

$$31 \leq 85 \quad T$$

$$13 \leq 85 \quad T$$

$$45 \leq 85 \quad T$$

$$103 \leq 85 \quad F$$

$$a[j] \geq a[\text{pivot}]$$

$$99 \geq 85 \quad T$$

$$26 \geq 85 \quad F$$

since $i < j$ ($5 < 7$)

swap 103 and 26

0 1 2 3 4 5 6 7 8 $i=5$
 $j=7$

85 76 31 13 45 26 -5 103 99

$$a[i] \leq a[\text{pivot}]$$

$$26 \leq 85 \quad T$$

$$a[j] \geq a[\text{pivot}]$$

$$-5 \leq 85 \quad T \quad (i=7)$$

$$103 \geq 85 \quad T$$

$$(j=6)$$

$$103 \leq 85 \quad F$$

$$i > j$$

$$(7 > 6)$$

swap pivot (85) $\leftrightarrow$ a[j]

-5 76 31 13 45 26 85 103 99

i j i j j j

first element is -5

pivot -5

$$a[i] \leq a[\text{pivot}]$$

$$-5 \leq -5 \quad T$$

$$76 \leq -5 \quad F$$

$$a[j] \geq a[\text{pivot}]$$

$$26 \geq -5 \quad T$$

$$45 \geq -5 \quad T$$

$$13 \geq -5 \quad T$$

$$31 \geq -5 \quad T$$

$$76 \geq -5 \quad T$$

$$-5 \geq -5 \quad T$$

-5 26 31 13 45 26 85 103 99

76	31	13	45	26
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pivot = 76
i = 76

$$a[i] \leq a[\text{pivot}]$$

$$a[j] \geq a[\text{pivot}]$$

$$76 \leq 76 \text{ T}$$

$$26 \geq 76 \text{ F}$$

$$36 \leq 76$$

$$13 \leq 76$$

$$45 \leq 76$$

$$26 \leq 76$$

$$i = 6$$

$$j = 5$$

$$i > j$$

swap 76 and 26

26	31	13	45	76
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pivot = 26

i = 26

$$a[i] \leq a[\text{pivot}]$$

$$a[j] \geq a[\text{pivot}]$$

$$26 \leq 26 \text{ T}$$

$$45 \geq 26 \text{ T}$$

$$31 \leq 26 \text{ F}$$

$$13 \geq 26 \text{ F}$$

$$i = 1 \quad j = 4$$

$$i < j$$

swap 13 & 31

26	13	31	45
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$$a[i] \leq a[\text{pivot}]$$

$$a[j] \geq a[\text{pivot}]$$

$$26 \leq 26 \text{ T}$$

$$45 \geq 26 \text{ T}$$

$$13 \leq 26 \text{ T}$$

$$31 \geq 26 \text{ T}$$

$$31 \leq 26 \text{ F}$$

$$13 \geq 26 \text{ F}$$

swap 13 & 26

13	26	31	45
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Then

103	99
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$$a[i] \leq a[\text{pivot}]$$

pivot = 103

$$103 \leq 103 \text{ T}$$

$$a[j] \geq a[\text{pivot}]$$

$$i = 103$$

$$99 \leq 103 \text{ F}$$

$$99 \geq 103 \text{ F}$$

$$i = 2 \quad j = 1$$

$$i > j$$

swap 103 & 99

so final sorted array

-5	13	26	31	45	76	85	99	103
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$i < j$

swap $a[i]$ & $a[j]$

$i > j$

swap
pivot &
 $a[j]$

i	0	1	2	3	4	5	6	7	8
	350	790	400	200	100	96	900	890	760
	i	i				j	j	j	j

① pivot is 350
 $i = 350$

$a[i] \leq a[\text{pivot}]$

$350 \leq 350$ T

$790 \leq 350$ F

350	96	100	200	400	790	900
	i	j		i		

760	890
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$a[j] \geq a[\text{pivot}]$

$760 \geq 350$ T

$890 \geq 350$ T

$900 \geq 350$ T

$96 \geq 350$ F

$100 \leq 350$ T

$200 \leq 350$ T

$400 \leq 350$ F

$400 \geq 350$ T

$200 \geq 350$ F

$j = 3$

$i = 4$

$a[j]$ & pivot

$i = 1$ $j = 5$

$i < j$

swap $a[i]$ & $a[j]$
790 & 96

350	96	400	200	100	790	900	890	760
i	i	i		j	j	j	j	j

②

$a[i] \leq \text{pivot}$

$96 \leq 350$

$400 \leq 350$ F

$a[j] \geq \text{pivot}$

$790 \geq 350$ T

$100 \geq 350$ F

$i = 1$

$j = 4$

$i < j$

swap i & j

200	96	100	350	400	790	900	760 890	760
0	1	2	3	4	5	6	7	8

$$a[i] \leq a[pivot]$$

$$200 \leq 200 \quad T$$

$$96 \leq 200 \quad T$$

$$100 \leq 200 \quad T$$

$$350 \leq 200 \quad F$$

$$a[j] \geq a[pivot]$$

$$760 \geq 200 \quad T$$

$$890 \geq 200 \quad T$$

$$900 \geq 200 \quad T$$

$$790 \geq 200 \quad T$$

$$400 \geq 200 \quad T$$

$$350 \geq 200 \quad T$$

$$100 \geq 200 \quad F$$

$$j = 2$$

$$i = 3$$

$$pivot = 200$$

$$a[j] = 100$$

$$i > j$$

swap pivot & $a[j]$

100	96	200	350	400	790	900	890	760
0	1	2	3	4	5	6	7	8

$$pivot = 100$$

$$a[i] \leq a[pivot]$$

$$100 \leq 100 \quad T$$

$$96 \leq 100 \quad T$$

$$200 \leq 100 \quad F$$

$$a[j] \geq a[pivot]$$

$$760 \geq 100 \quad T$$

$$890 \geq 100 \quad T$$

$$900 \geq 100 \quad T$$

$$790 \geq 100 \quad T$$

$$400 \geq 100 \quad T$$

$$350 \geq 100 \quad T$$

$$200 \geq 100 \quad T$$

$$96 \geq 100 \quad F$$

$$i = 2$$

$$j = 1$$

$$i > j$$

swap pivot & $a[j]$

96	100	200	350	400	790	900	890	760
0	1	2	3	4	5	6	7	8

$$pivot = 300$$

$$a[i] \leq a[pivot]$$

$$350 \leq 350 \quad T$$

$$400 \leq 350 \quad F$$

$$a[j] \geq a[pivot]$$

$$760 \geq 350 \quad T$$

$$890 \geq 350 \quad T$$

$$900 \geq 350 \quad T$$

$$790 \geq 350 \quad T$$

$$400 \geq 350 \quad T$$

$$350 \geq 350 \quad T$$

swap $a[j]$ & pivot

96	100	200	350	400	790	900	890	760
				1	1			

$$pivot = 400$$

$$a[i] \leq a[\text{pivot}]$$

$$400 \leq 400 \quad T$$

$$790 \leq 400 \quad F$$

swap $a[j]$ & pivot

0	1	2	3
96	100	200	350

$$a[j] \geq a[\text{pivot}]$$

$$760 \geq 400$$

$$890 \geq 400$$

$$900 \geq 400$$

$$790 \geq 400$$

$$400 \geq 400$$

5	6	7	8
790	900	890	760
i	i		j

$$a[i] \leq a[\text{pivot}]$$

$$790 \leq 790 \quad T$$

$$900 \leq 790 \quad F$$

$$i < j$$

swap $a[i]$ & $a[j]$

$$a[j] \geq a[\text{pivot}]$$

$$760 \geq 790 \quad F$$

$$i = 6$$

$$j = 8$$

790	760	890	900
	j	i	

$$a[i] \leq a[\text{pivot}]$$

$$790 \leq 790 \quad T$$

$$760 \leq 790 \quad \uparrow$$

$$890 \leq 790 \quad F$$

$$i > j$$

swap $a[j]$ & pivot

760	790	890	900
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$$a[j] \geq a[\text{pivot}]$$

$$900 \geq 790 \quad T$$

$$890 \geq 790 \quad T \quad i = 7$$

$$760 \geq 790 \quad F \quad j = 6$$

final sorted array

96	100	200	350	400	760	790	890	900
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